



## Code-Layout, Readability and Reusability

|                      |                   |
|----------------------|-------------------|
| <b>Date</b>          | 28 September 2026 |
| <b>Team ID</b>       | SWTID-2026-6721   |
| <b>Project Name</b>  | FAQ Assistant API |
| <b>Maximum Marks</b> | 5 Marks           |

### Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

### Code Layout Checklist:

| S.No | Code Quality Parameter    | Description                                                   | Followed (Yes / No / Partial) | Remarks                                                                               |
|------|---------------------------|---------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------|
| 1    | Consistent Indentation    | Uniform spacing and indentation are used throughout the code. | YES                           | Consistent indentation is used throughout the Python code.                            |
| 2    | Proper File Structure     | Files and folders are logically organize                      | YES                           | Project files are organized logically for API development.                            |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly                       | YES                           | Variables such as question and answer clearly represent their purpose.                |
| 4    | Function / Method Names   | Functions are descriptively named                             | YES                           | Functions are named clearly according to their purpose.                               |
| 5    | Code Comments             | Inline and block comments explain log c                       | YES                           | Comments are used where necessary to explain the code.                                |
| 6    | Modular Design            | Code is split into reusable functions/modules                 | YES                           | The API is organized into reusable components for processing questions and responses. |

|   |                   |                                              |     |                                                                          |
|---|-------------------|----------------------------------------------|-----|--------------------------------------------------------------------------|
| 7 | No Redundant Code | Duplicate or unused code is removed          | YES | Unnecessary and duplicate code is avoided in the project.                |
| 8 | Error Handling    | Exceptions and errors are handled gracefully | YES | Invalid or empty questions are handled with appropriate error responses. |

**Reusable Components / Modules:**

| S.No | Component / Module Name | Language / Technology | Where Reused                                  | Reusability Level (High / Medium / Low) |
|------|-------------------------|-----------------------|-----------------------------------------------|-----------------------------------------|
| 1    | FAQ API Endpoint        | Python / Fast API     | Used for handling user questions              | High                                    |
| 2    | Gemini AI Integration   | Python / Gemini AI    | Used for generating answers                   | High                                    |
| 3    | JSON Request & Response | JSON                  | Used for API communication                    | High                                    |
| 4    | Input Validation        | Python / FastAPI      | Used for validating user questions            | Medium                                  |
| 5    | Error Handling          | Python                | Used for handling invalid requests and errors | Medium                                  |

**Overall Code Quality Assessment:**

| Aspect                   | Rating (1-5) | Comments                                                         |
|--------------------------|--------------|------------------------------------------------------------------|
| Code Layout & Structure  | 5            | The project has a clear and organized API structure.             |
| Readability              | 5            | The code is simple and easy to understand                        |
| Reusability              | 4            | API components can be reused in other applications.              |
| Documentation / Comments | 4            | Proper documentation and necessary comments are provided.        |
| Overall Score            | 5            | The project has good code quality, readability, and reusability. |